

12 The specific heat capacity of different substances is shown.

substance	specific heat capacity / $\text{J g}^{-1} \text{ } ^\circ\text{C}$
air	1.0
hydrogen gas	14.3
water (liquid)	4.2
water (steam)	2.1

Which statement is correct?

- A** Air is a more stable environment than water because it is more resistant to changes in temperature.
- B** It takes more energy to raise the temperature of hydrogen gas than it does to raise the temperature of water.
- C** A specific heat capacity of $4.2 \text{ J g}^{-1} \text{ } ^\circ\text{C}$ means that it takes 4.2 J of energy to vaporise 1.0 g of liquid water.
- D** There are more hydrogen bonds between water molecules in a gas than between water molecules in a liquid.